

Priyanka Chatterjee

AI / Backend Engineer · 9,300 ★ Open-Source Contributor · Python · RAG ·
MCP · 3.5+ Years
Kolkata, West Bengal, India (open to remote)

+91 8617262536
priyankachatterjee9075@gmail.com
linkedin.com/in/priyanka-9075
github.com/Priyanka-Chatterjee-2000

SUMMARY

Accomplished AI / Backend Engineer with 3+ years building production AI systems in **Python** — **RAG** pipelines, **multi-agent LLM** workflows, and **Model Context Protocol (MCP)** servers across **AWS**, Azure, and GCP. Core contributor to four open-source products at Turbot (120+ merged pull requests, 9,300+ GitHub stars), shipping a Steampipe **MCP** server that gives **LLM** agents standardized, real-time access to enterprise data.

TECHNICAL SKILLS

Languages: Python, Go, TypeScript, SQL, Bash

AI & Agentic Engineering: LLM Application Development, Agentic Workflows, Multi-Agent Systems, Retrieval-Augmented Generation (RAG), Prompt Engineering, Model Context Protocol (MCP), OpenAI Agents SDK, Pydantic AI, Logfire (LLM observability), Claude Code

Backend & APIs: Microservices, Event-Driven Architecture, REST, Serverless (AWS Lambda)

Cloud & DevOps: AWS (Lambda, EC2, S3), Azure, GCP, Docker, Kubernetes (working knowledge), GitHub Actions

Data & Databases: PostgreSQL, DuckDB, SQL, pandas, Parquet

Testing & Quality: Unit Testing, Integration Testing, Code Review

Core CS: Data Structures & Algorithms, Object-Oriented Design, System Design

WORK EXPERIENCE

Turbot HQ, Inc. Remote

August 2022 – April 2026

Software Engineer I

- Core contributor across all four Turbot open-source products (Steampipe, Powerpipe, Flowpipe, Tailpipe) — 120+ merged pull requests and 4,200+ commits across 79 repositories, with 9,300+ combined GitHub stars.
- Built the Steampipe **Model Context Protocol (MCP)** server end to end in **Python**, giving **LLM** agents standardized, real-time access to enterprise cloud and security data for retrieval and analysis.
- Designed a **multi-agent** workflow on the OpenAI Agents SDK to orchestrate **LLM** agents that generate cloud preventive-policy commands, validate them in a live test environment, self-correct failures, and score accounts against an internal security metric.
- Built a **RAG**-based internal Pydantic AI agent to resolve cross-team queries by ingesting the documentation and code graph for all internal repos into S3 vector storage; deployed it to **AWS** Lambda and integrated it with the Slack Bot API as the UI for easy developer access, resolving onboarding and cross-team queries instantly.
- Authored an **LLM**-driven Claude Code skill automating code refactoring, unit test generation, deployment, and PR creation; used it to migrate 35+ Azure cloud-governance modules from polling to push-based eventing, cutting resource-discovery latency 70-90%.
- Engineered core features for Tailpipe, a Go-based observability CLI on DuckDB + Parquet, lifting log-processing throughput 30% to 1,000+ logs per second by chunking and batching file reads and streaming writes to DuckDB; reduced memory use to eliminate out-of-memory failures.

- Optimized complex **PostgreSQL** queries powering customer compliance audits — rewriting **AWS** IAM policy-evaluation queries with jsonb array elements, multi-stage CTEs, and condition-aware filters — improving query performance 40% and eliminating a class of false-positive findings across 10,000+ resources.
- Resolved a production incident by diagnosing a Go service stuck in a **Kubernetes** CrashLoopBackOff, collaborating with the platform team to investigate pod logs and status and restore service.

PROJECTS

General-Purpose AI Chatbot / Agent (Model Context Protocol) *(Jun 2025 – Aug 2025)*

Agentic chatbot/agent framework built with Pydantic AI featuring configurable external **MCP** servers via a persistent MongoDB-backed registry, JSON config loader with strict input validation, per-server failure isolation, native tool calling, mid-conversation multi-model switching (OpenAI, Gemini, Anthropic, Mistral), conversation-history persistence, sync + streaming output, prompt-injection safeguards, and Logfire observability.

CIS Benchmark → Compliance Dashboard Generator

Python CLI that automates compliance-dashboard authoring for a multi-cloud governance product. Ingests CIS benchmark XLSX workbooks into ready-to-deploy dashboard config plus markdown docs. OOP resource model, custom HCL pretty-printer, openpyxl + **pandas**, distributed as a pip-installable package.

EDUCATION

University of Engineering & Management (UEM) Kolkata, WB, IN

August 2019 – May 2023

Bachelor of Technology in Computer Science and Engineering

CGPA: 9.28